Ethanol Blending in Gasoline – Japan

July 2026

U.S. GRAINS & BIOPRODUCTS COUNCIL

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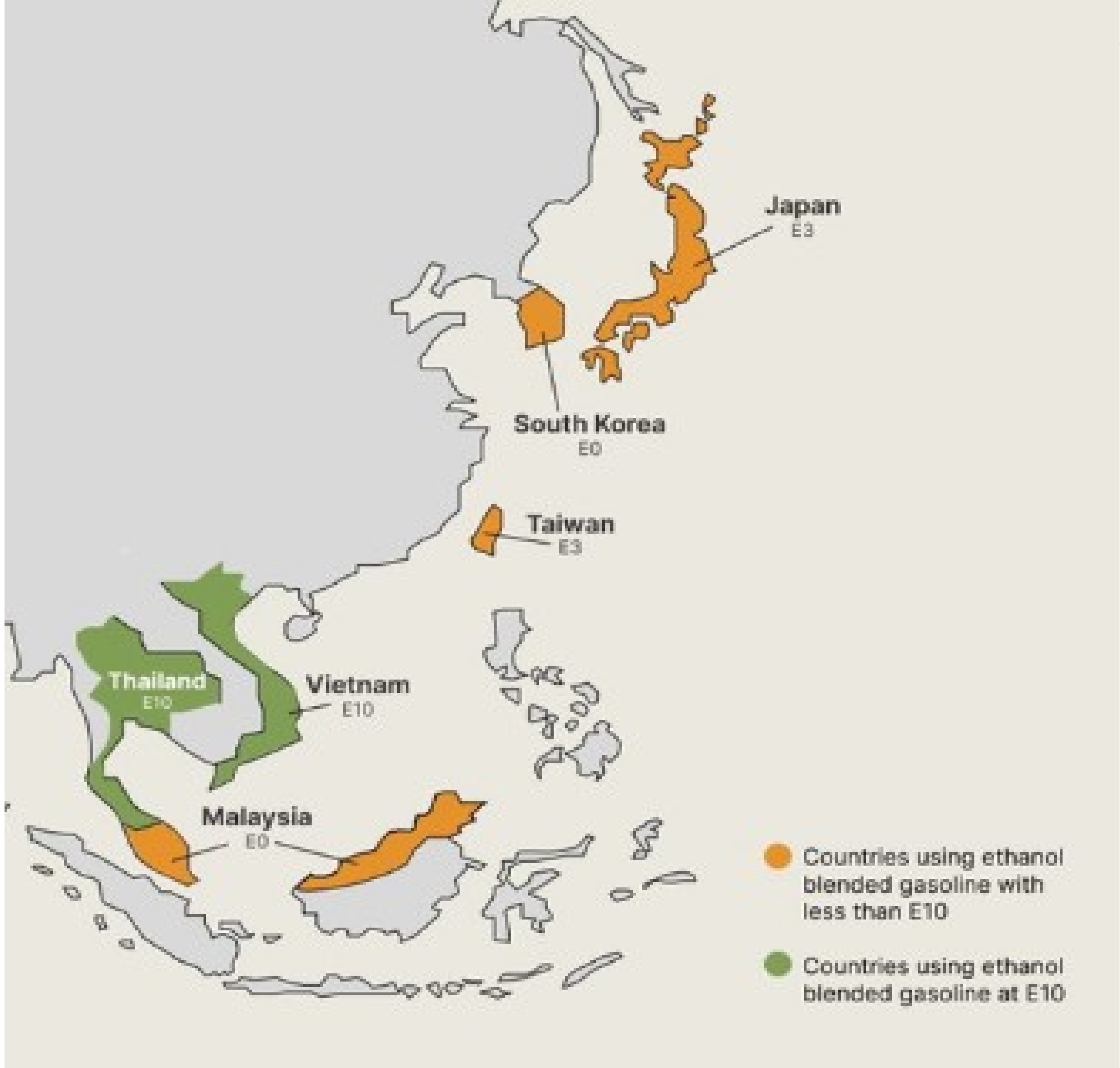
Ethanol Blending in Asia

There are important fuel quality and environmental impact of vehicle emission challenges in these regions.

- The use of ethanol improves gasoline quality and creates flexibility in gasoline production.
- Ethanol use is a cost-effective way to increase gasoline octane and to replace more expensive gasoline components.
- Ethanol contributes to transport decarbonization and air quality improvement.
- There are opportunities across Asia to increase the ethanol blend level and implement new policies on the use of gasoline-ethanol blends.



Seven countries with potential and additional use of ethanol were studied: 1) Gasoline market profiles; 2) Optimization of gasoline blends with ethanol and, 3) Environmental impact of gasolines blended with ethanol.

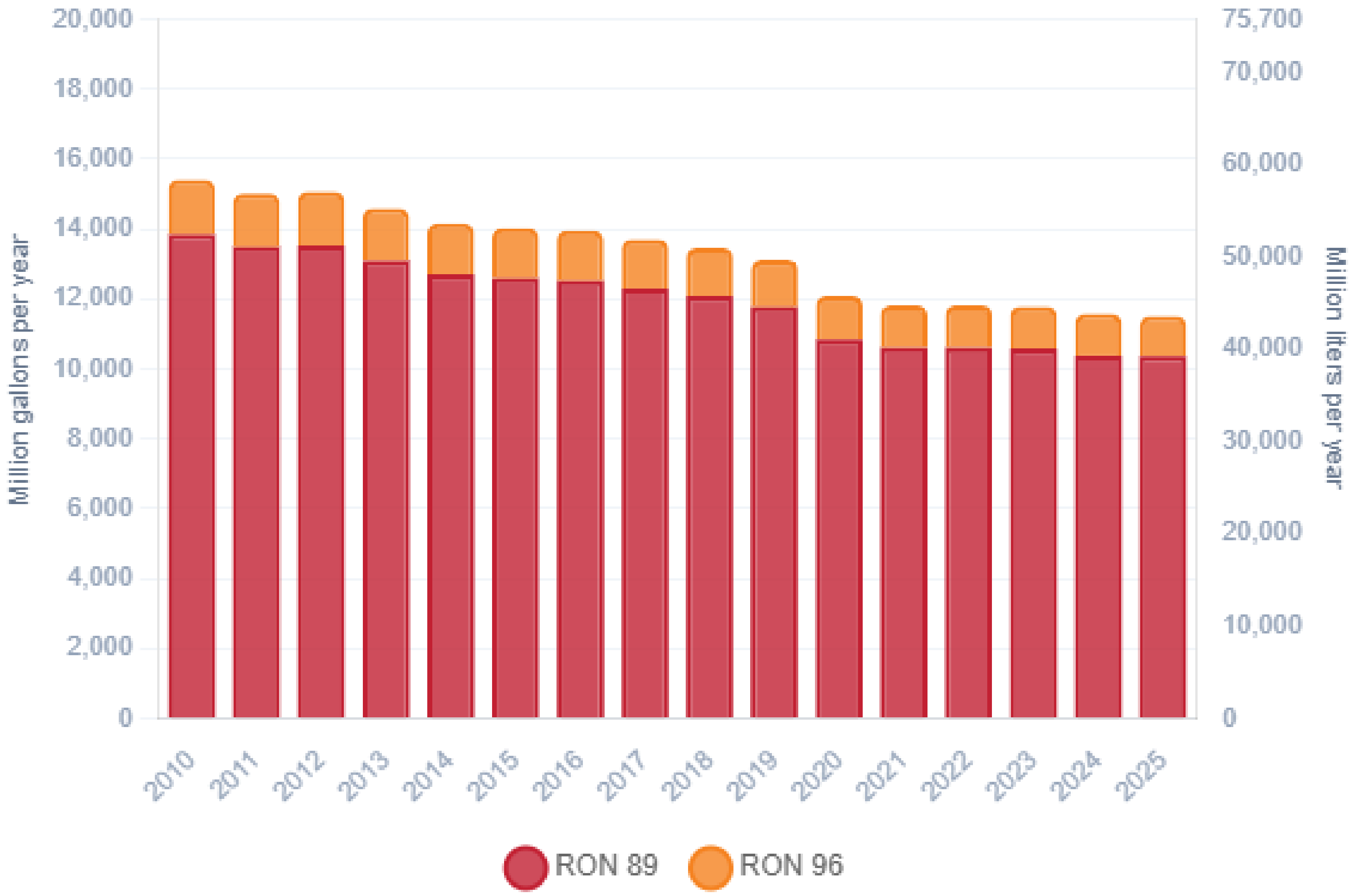
Ethanol Gasoline Blending in Japan

In 2025, gasoline consumption in Japan was 43,581 million liters (11,513 million gallons) and growth rate was -1.9%. Gasoline production and net imports were 42,543 and 877 million liters (11,107 and 231 million gallons) correspondingly. Japan complies with the JIS standards and offers two main grades: RON 89 and RON 96) with an average octane of 89.7 RON.

Japan currently allows E3. Last year's consumption was 745 million liters (197 million gallons) equivalent to 1.7%, primarily as the component ethyl tert-butyl ether (ETBE). There is no local production. Ethanol is imported as a blendstock for ETBE production and as ETBE. The Ministry of Economy, Trade and Industry (METI) aiming for E10 (10% ethanol) by 2030 and E20 (20% ethanol) by 2040. This initiative requires infrastructure improvements, as most existing gas stations are only equipped for the E3 blend.

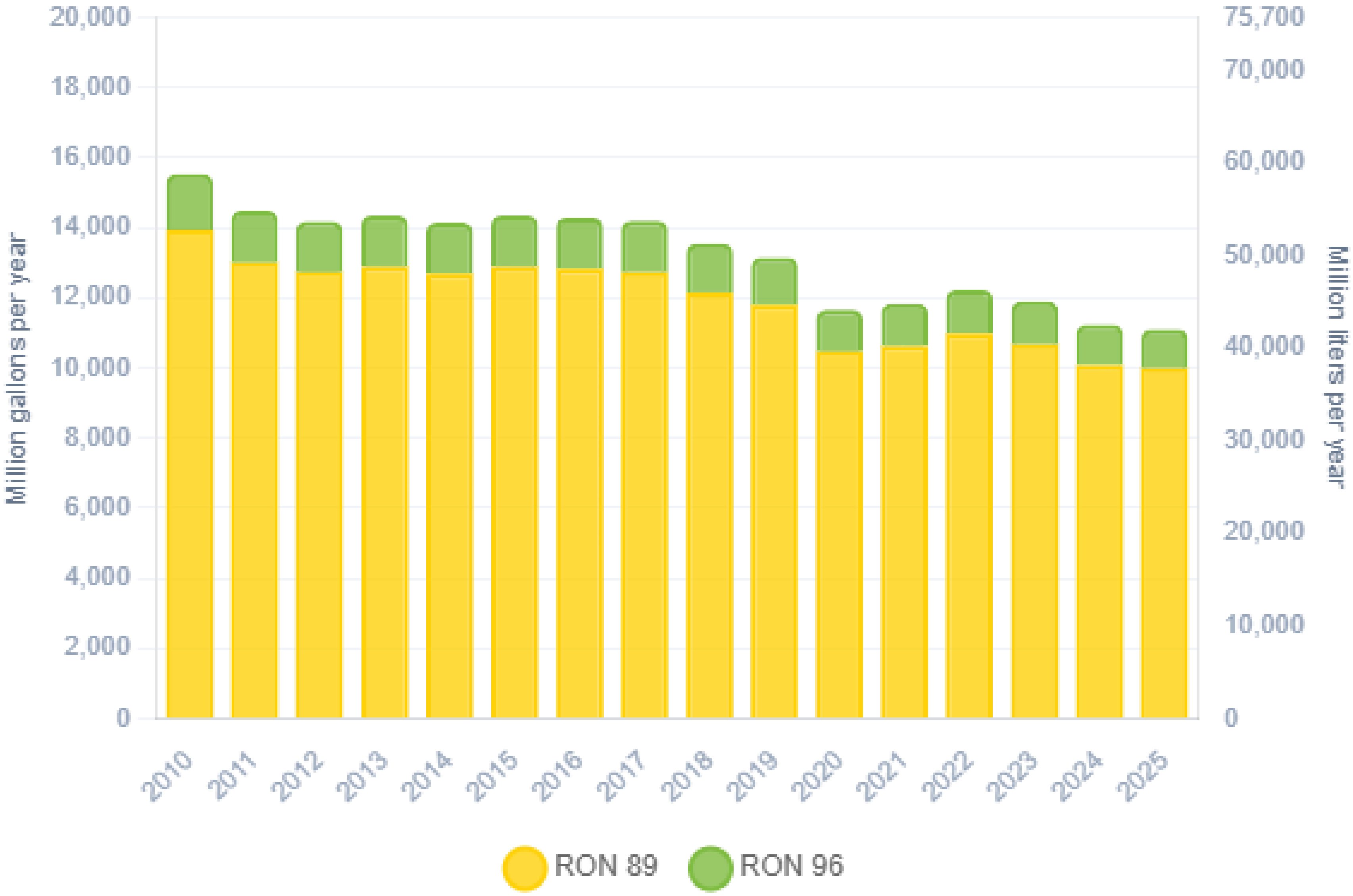
Source: METI, USDA

Gasoline Demand in Japan



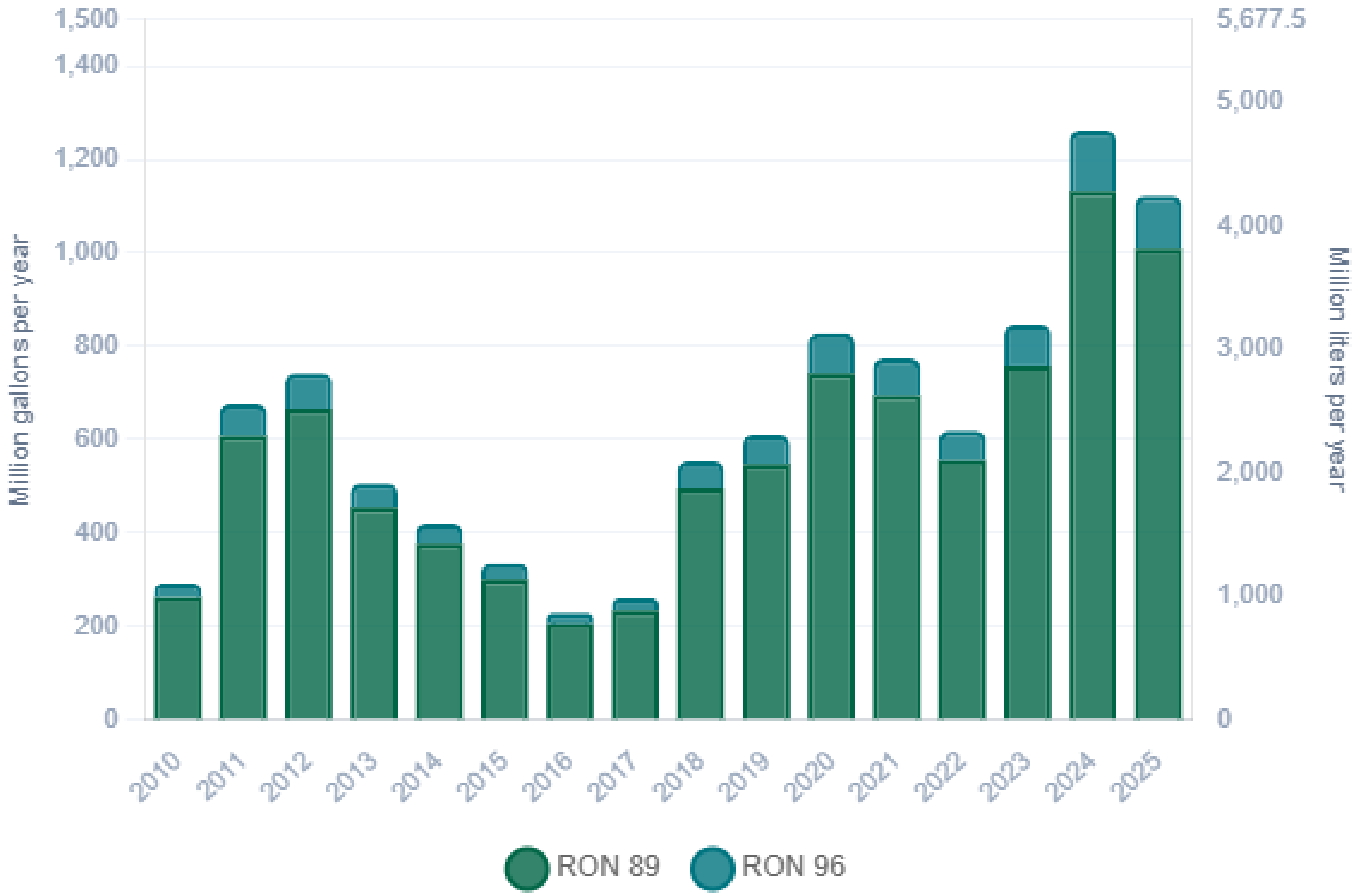
Source: METI

Gasoline Production in Japan



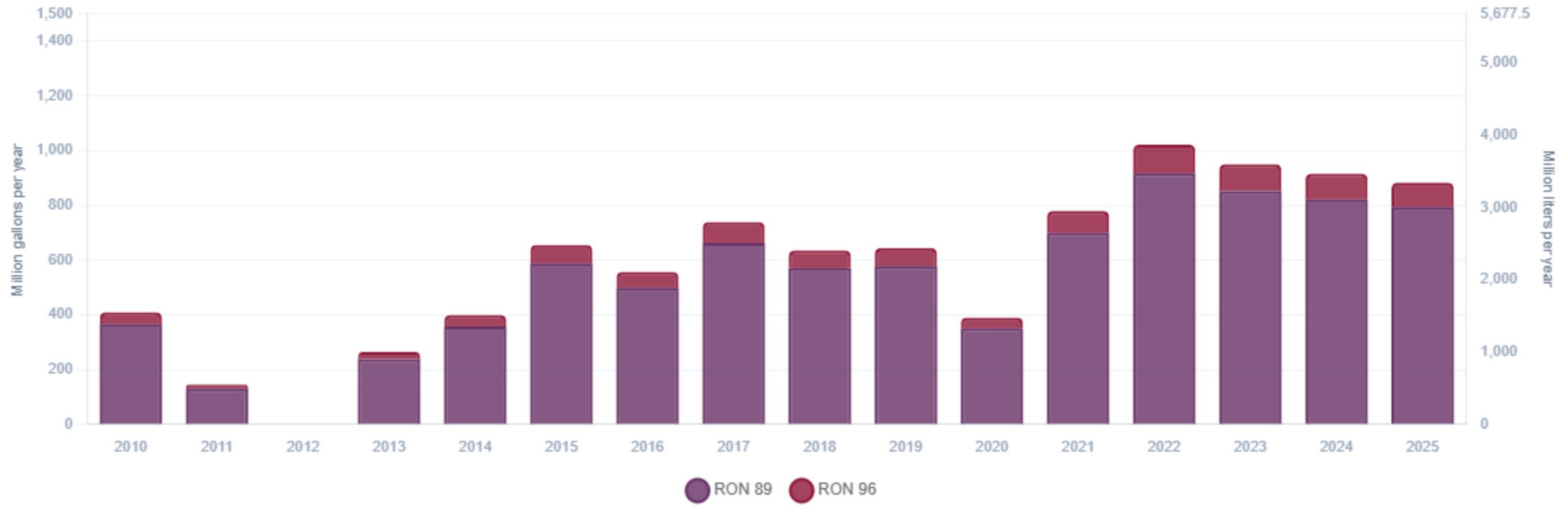
Source: METI

Gasoline Imports in Japan



Source: METI

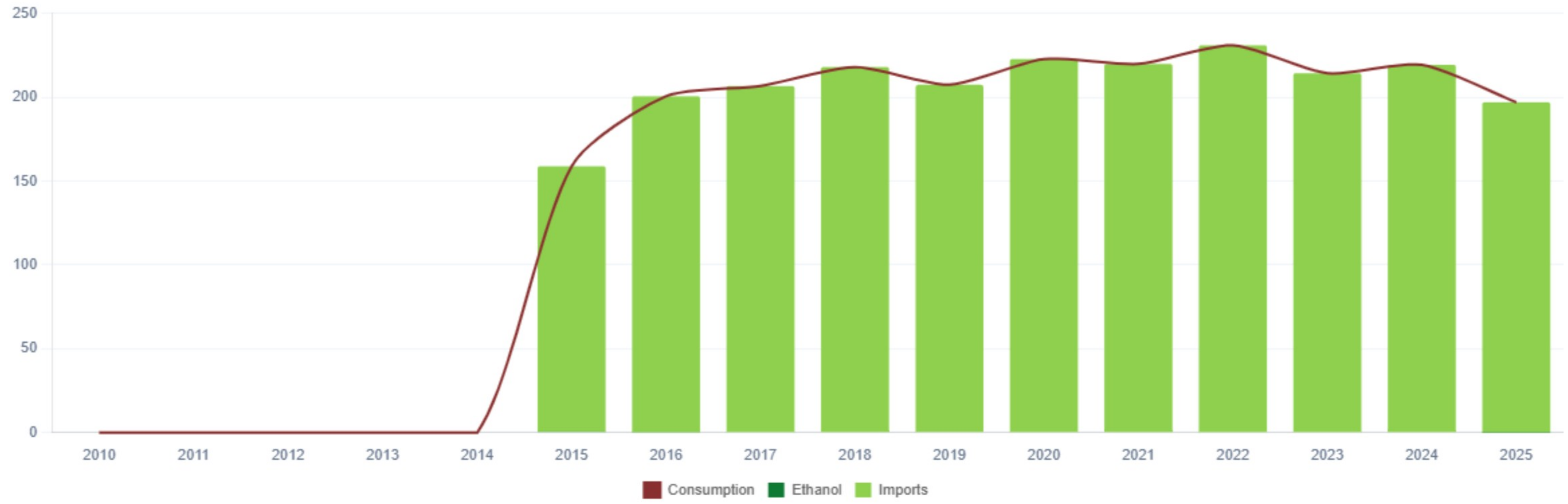
Gasoline Exports in Japan



Source: METI

Ethanol Balance in Japan

There is no recent local production of ethanol. Ethanol is supplied by imports of ethanol as a blendstock for ETBE production or as ETBE



Source: USDA, METI

Gasoline Quality in Japan

Name	JIS K 2202: 2023 - Law on the Quality Control of Gasoline and Other Fuels							
Implementation Date	2023							
Applicability	Japan							
Grade	Regular		Premium		E10 Regular *		E10 Premium *	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	89.0		95.0		90.0		96.0	
MON								
AKI								
Benzene (%vol)		1.0		1.0		1.0		1.0
Aromatics (% vol)		-		-		-		-
Olefins (%vol)		-		-		-		-
Lead (g/L)		-		-		-		-
Manganese (mg/L)		-		-		-		-
Sulfur (ppm)		10		10		10		10
Oxygen (%w)		1.3		1.3		3.5		3.5
Ethanol (%vol)		3.0		3.0	3	10.0	3.0	10.0
MTBE (%vol)		7.0		7.0				
PVR 37.8°C (psi)	6.4	11.3	6.4	11.3	6.4	11.3	6.4	11.3
Carbon Intensity (gCOe/MJ)	0	88.74	-	88.74	0	88.74	-	88.74
Kerosene (%vol)	-	4.0	-	4.0	-	4.0	-	4.0

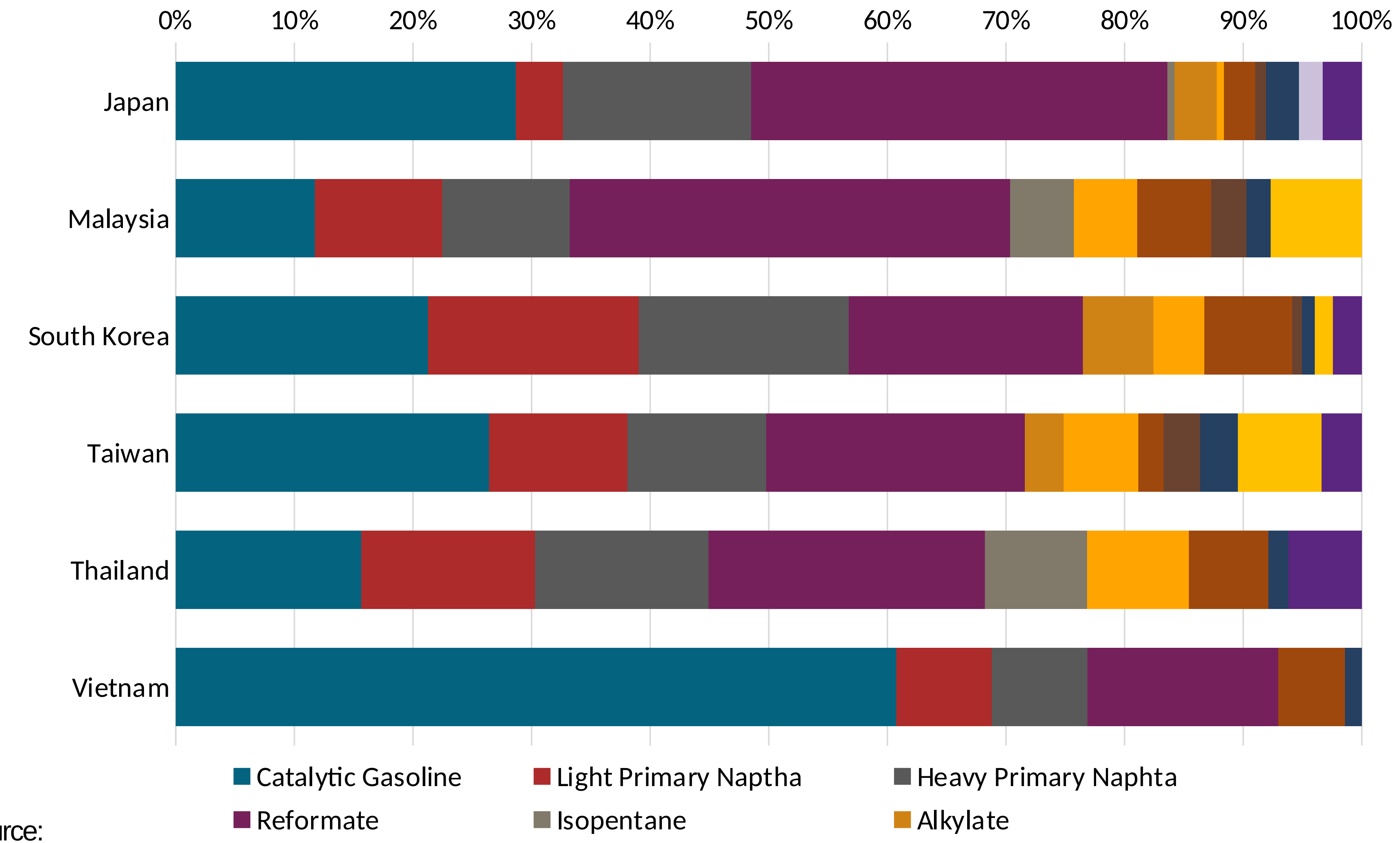
*Not Mandatory. A 55% Carbon Intensity Reduction target has been established and will be revised once LCA has been completed

Source: METI

Gasoline Component Blending in Asia

Gasoline is a blend of a base gasoline and other components. Each component provides different properties to the final blend, for example, isomerates, alkylates and butanes increase the octane. The components commonly used in Asia are:

- Catalytic Gasoline
- Light Primary Naptha
- Heavy Primary Naphta
- Reformate
- Isopentane
- Alkylate
- Isomerate
- Hydrocracked Gasoline
- Coker Naphta
- n-Butane
- MTBE
- ETBE
- TAME
- Aromatics



Source: Faro90

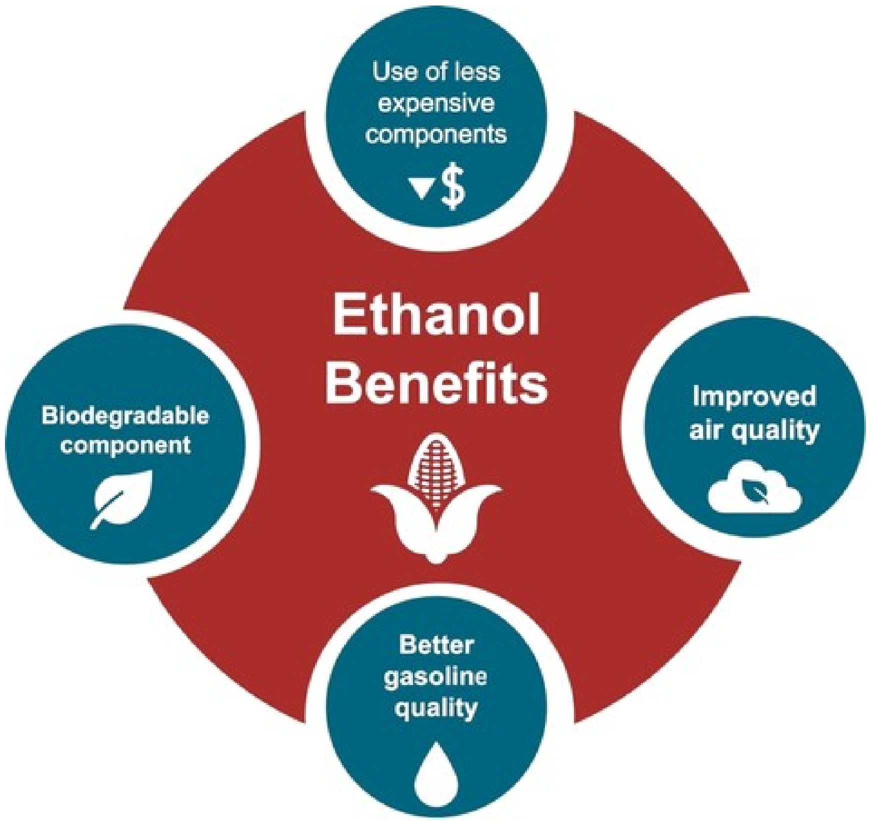
Gasoline Blending Optimization

In some parts of the world, ethanol is added to gasoline as a blending component.The advantages of ethanol include that it is a renewable fuel made of biomass; that it is an octane booster that helps to dilute sulfur; and that it allows the fulfillment of environmental objectives.To determine the optimal components to be blended with ethanol, a **blending model** was used. This model selects the components to add in the gasoline/ethanol blend based on:

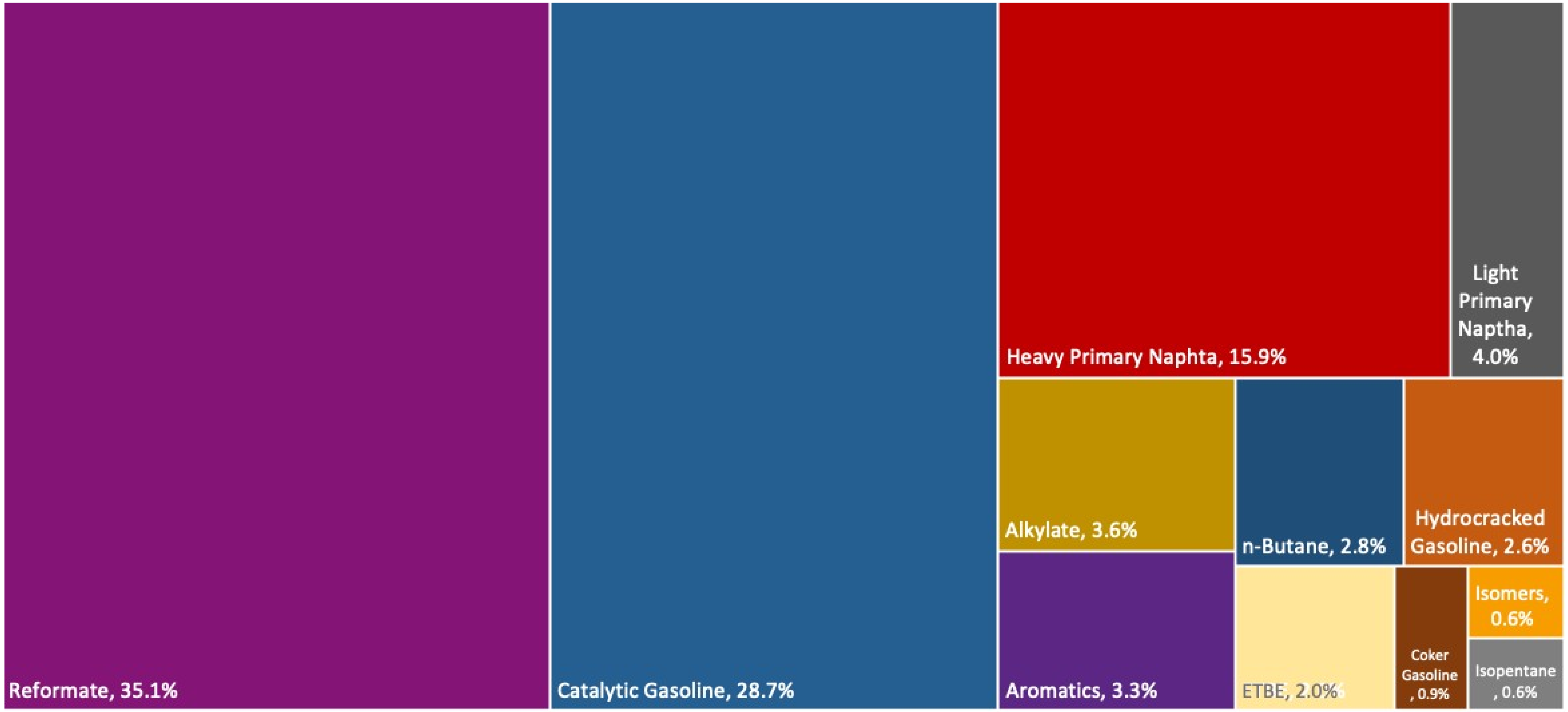
- Components prices,
- Properties each component affects,
- Quality parameters by country, and
- Component availability by country.

Through iterations, the model obtains the %v/v of the components to be blended with 10%, 15%, 20%, 25% and 30% of ethanol, in such a way that the final blend complies with the required properties of a finished gasoline by country.

The blending model uses gasoline component spot average prices 2025 and provides fuel prices that do not include country distribution costs, local taxes and subsidies and import or gas station margins.

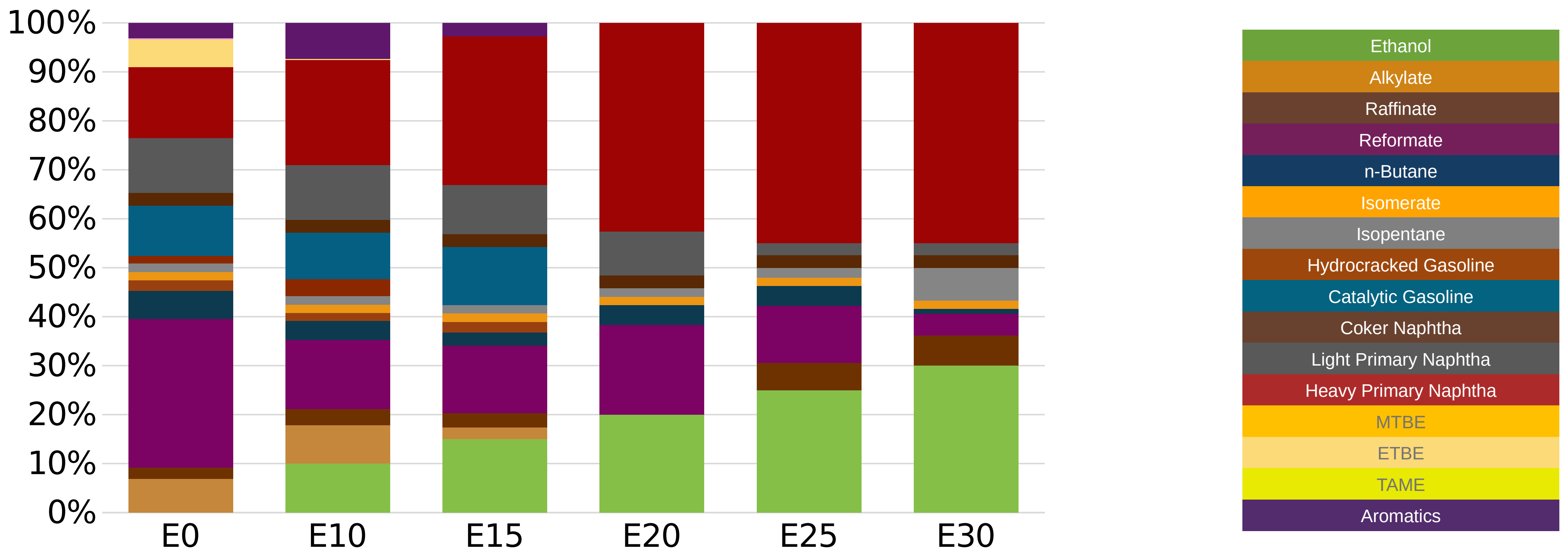


Japan Available Blending Components



Source:
Faro90

Japan – Gasoline 1 – Constant Octane

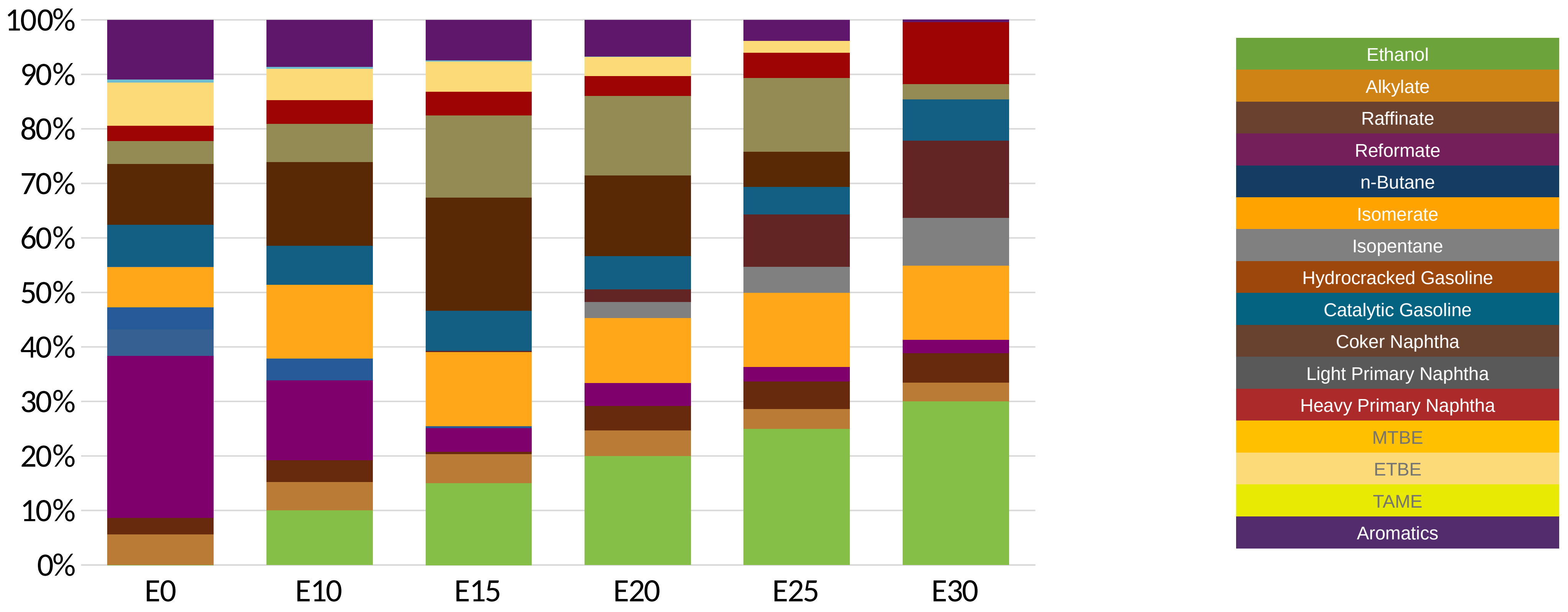


Octane (RON)	89.0	89.0	89.0	89.0	90.0	90.7
Price (US$/gal)	$2.01	$1.76	$1.61	$1.46	$1.40	$1.32

Average CIF 2025 Prices

Source: Faro90

Japan - Gasoline 2 - Constant Octane

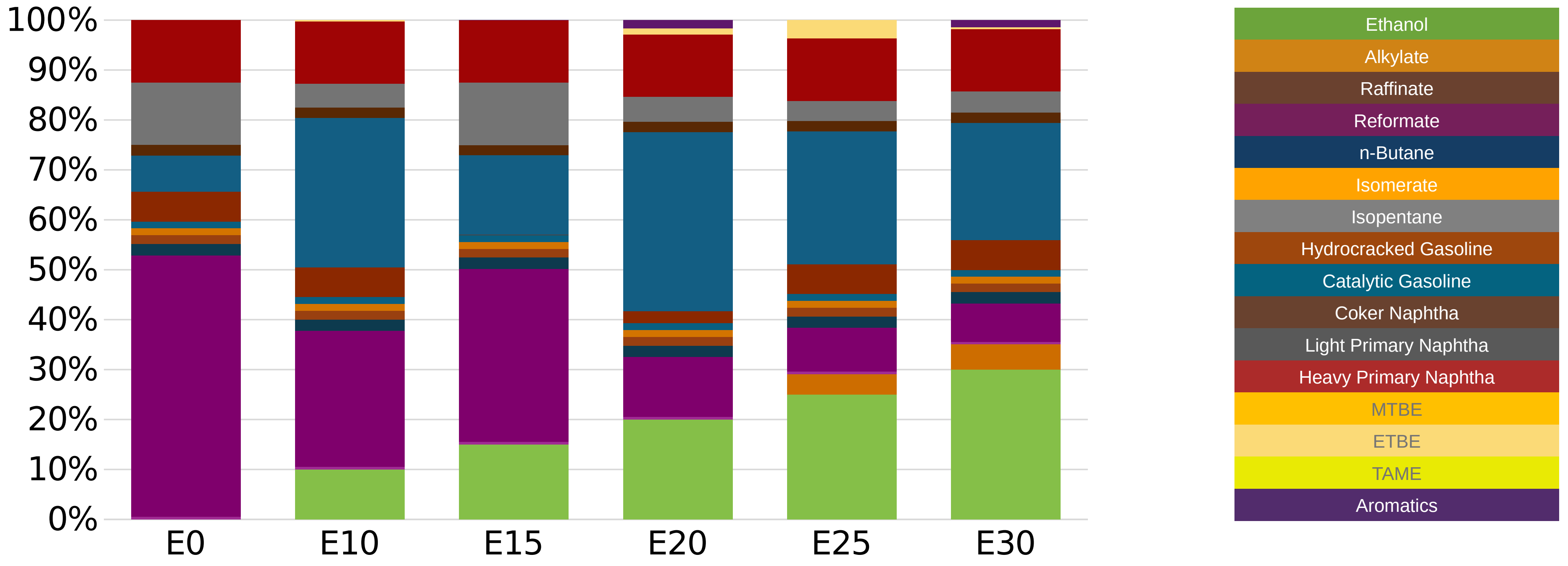


Octane (RON)	96.0	96.0	96.0	96.0	96.0	96.0
Price (US$/gal)	$2.20	$2.13	$2.05	$1.97	$1.88	$1.77

Average CIF 2025 Prices

Source: Faro90

Japan – Gasoline 1 – Increased Octane

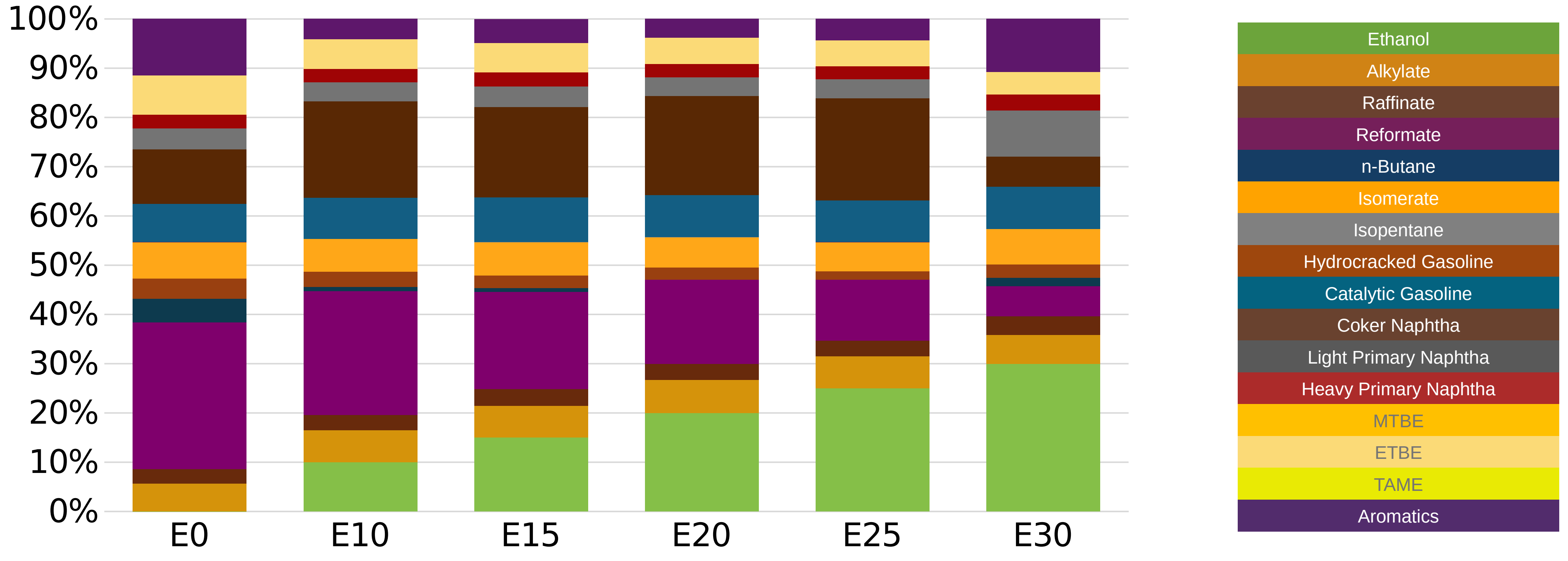


Octane (RON)	89.2	93.3	95.2	97.1	98.8	100.0
Price (US$/gal)	$2.01	$2.01	$2.01	$2.01	$2.01	$2.01

Average CIF 2025 Prices

Source: Faro90

Japan – Gasoline 2 – Increased Octane



Octane (RON)	96.0	97.6	99.1	100.5	102.0	103.2
Price (US$/gal)	$2.20	$2.20	$2.20	$2.20	$2.20	$2.20

Average CIF 2025 Prices

Source: Faro90

Vehicle Emission Impact for Ethanol Gasoline Blending

The model used in this analysis takes as a reference the **International Vehicle Emissions Model (IVE).**

The model uses the Base Emission Rates from IVE model, as well as its Adjustment Factors based on:

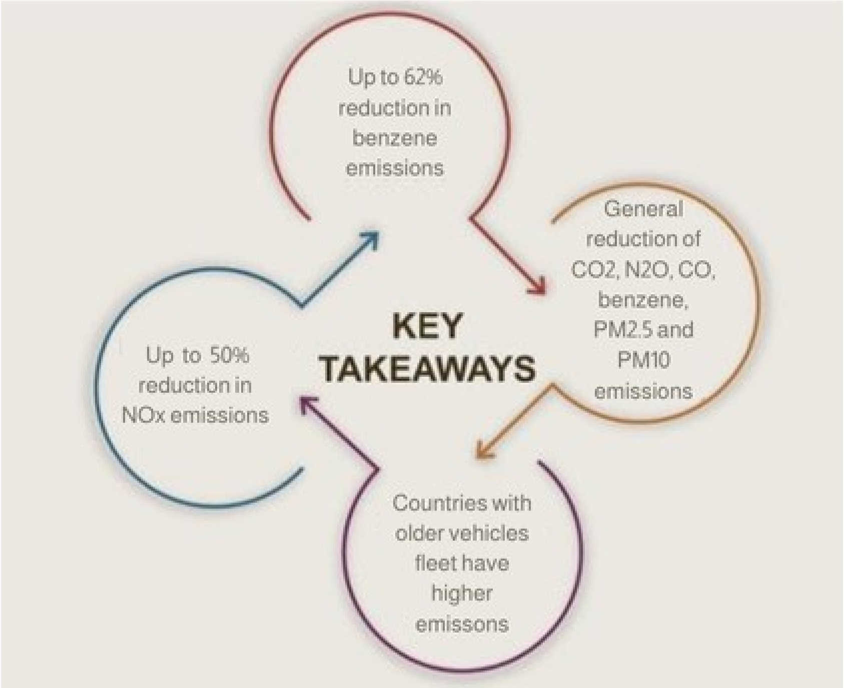
- Vehicle technology (cars, trucks, buses, motorcycles),
- Vehicle fleet average age,
- Average traveled distance per vehicle by country, as well as
- Geographical and climatic conditions (altitude, humidity, temperature).

Emissions of criteria pollutants, toxic pollutants, and greenhouse gases (GHG) were calculated and calibrated with emission inventories, using real gasoline quality data. The reduction rates for gasoline/ethanol blends were obtained from various sources (IPCC, US Grains, among others).

Emission estimations for different pollutants for gasoline and gasoline/ethanol blends (10%, 15%, 20%, 25% and 30% ethanol) were determined using the IVE Model. A comparison between the results and the European (Euro 6) requirements is made. Results are also compared with real emissions of the United States vehicle fleet*.

Source: Bureau of transportation statistics.

Main Results



Gasoline Vehicle Fleet - Japan

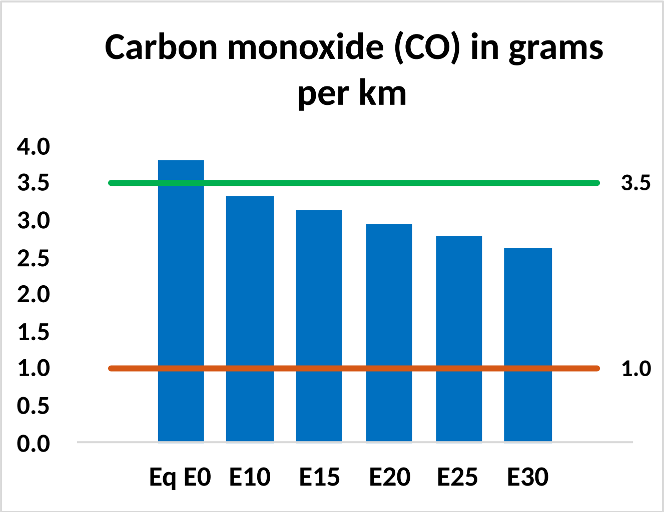
Type	Weight	Engine	Exhaust	Recovery	Age	No. Vehicles
Auto/Sml Truck	Light	Carburat	3-Ways	PCV Tank	>8 years	191,360
Auto/Sml Truck	Light	Single FI	3-Ways	PCV Tank	>8 years	819,341
Auto/Sml Truck	Light	Multi FI	3-Ways	PCV Tank	>8 years	4,375,203
Auto/Sml Truck	Light	Multi FI	Euro III	PCV Tank	>8 years	5,105,687
Auto/Sml Truck	Light	Multi FI	Euro IV	PCV Tank	>8 years	8,486,532
Auto/Sml Truck	Light	Multi FI	Euro V	PCV Tank	>8 years	2,778,754
Auto/Sml Truck	Light	Multi FI	Euro 6	PCV Tank	>8 years	6,882,393
Auto/Sml Truck	Light	Multi FI	Euro 6	PCV Tank	4-8 years	8,213,163
Auto/Sml Truck	Light	Multi FI	Euro 6	PCV Tank	<4 years	6,372,261
Auto/Sml Truck	Light	Multi FI	Hybrid	PCV Tank	All	13,710,087
Auto/Sml Truck	Light	Electric			All	84,483
Auto/Sml Truck	Medium	Carburat or	3-Ways	PCV Tank	>8 years	179,914
Auto/Sml Truck	Medium	Single FI	3-Ways	PCV Tank	>8 years	1,038,575
Auto/Sml Truck	Medium	Multi FI	3-Ways	PCV Tank	>8 years	1,069,752
Auto/Sml Truck	Medium	Multi FI	Euro III	PCV Tank	>8 years	1,025,436
Auto/Sml Truck	Medium	Multi FI	Euro IV	PCV Tank	>8 years	2,063,688
Auto/Sml Truck	Medium	Multi FI	Euro V	PCV Tank	>8 years	844,450
Auto/Sml Truck	Medium	Multi FI	Euro 6	PCV Tank	>8 years	1,954,564
Auto/Sml Truck	Medium	Multi FI	Euro 6	PCV Tank	4-8 years	2,176,458
Auto/Sml Truck	Medium	Multi FI	Euro 6	PCV Tank	<4 years	2,037,277
Auto/Sml Truck	Medium	Multi FI	Hybrid	PCV Tank	All	3,929,919
Auto/Sml Truck	Medium	Electric			All	294,731
Sml Engines	Light	4-Cycle	None	PCV	>8 years	237,867
Sml Engines	Light	4-Cycle	Euro III	PCV	>8 years	346,124
Sml Engines	Light	4-Cycle	Euro IV	PCV	>8 years	882,856
Sml Engines	Light	4-Cycle	Euro V	PCV	>8 years	278,271
Sml Engines	Light	4-Cycle	Euro 6	PCV	>8 years	702,464
Sml Engines	Light	4-Cycle	Euro 6	PCV	4-8 years	694,349
Sml Engines	Light	4-Cycle	Euro 6	PCV	<4 years	921,228
Sml Engines	Light	4-Cycle	Hybrid	PCV	All	178,979
Sml Engines	Light	Electric			All	189,122
Truck	Heavy	Carburat	3-Ways	PCV	>8 years	54,410
Truck	Heavy	Fuel-Injection	3-Ways	PCV	>8 years	637,153
Truck	Heavy	Fuel-Injection	Euro III	PCV	>8 years	164,112
Truck	Heavy	Fuel-Injection	Euro IV	PCV	>8 years	411,358
Truck	Heavy	Fuel-Injection	Euro V	PCV	>8 years	198,040
Truck	Heavy	Fuel-Injection	Euro 6	PCV	>8 years	863,691
Truck	Heavy	Fuel-Injection	Euro 6	PCV	4-8 years	1,320,080
Truck	Heavy	Fuel-Injection	Euro 6	PCV	<4 years	1,174,293
Truck	Heavy	Fuel-Injection	Hybrid	PCV	All	284,413
Truck	Heavy	Electric			All	321,973

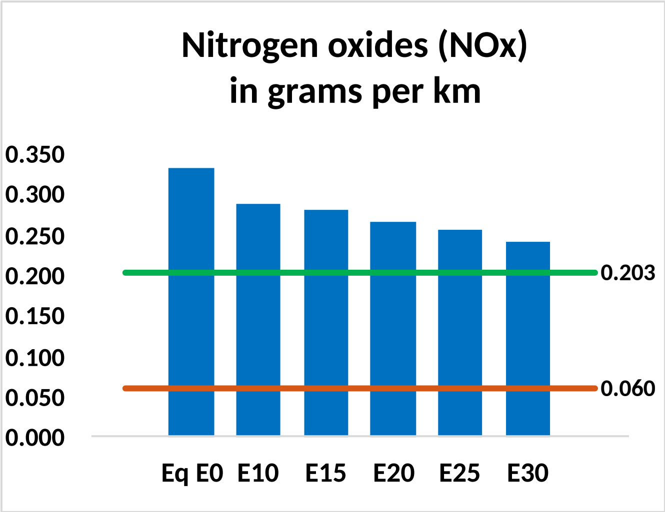
Vehicle Fleet: 89,079,267
Gasoline Vehicle Fleet: 81,722, 132

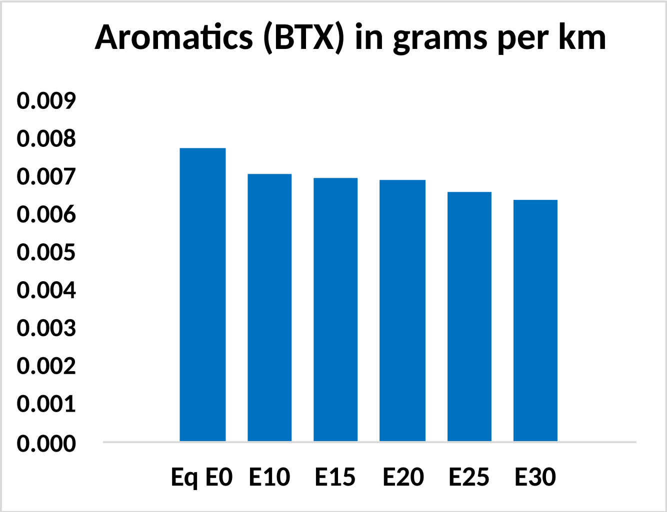
Source: AIRIA

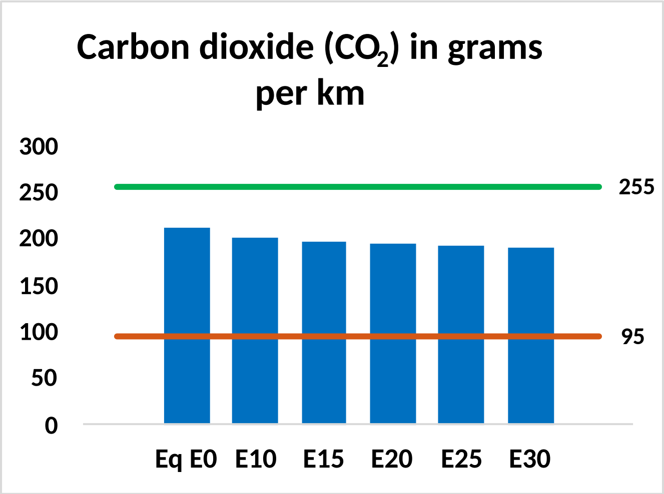
Japan – Vehicular Emissions

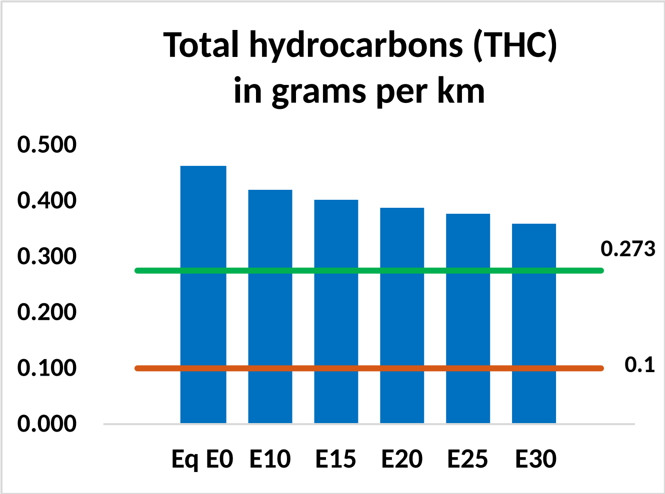
TIER III BIN 125 United States
Euro 6 Standard

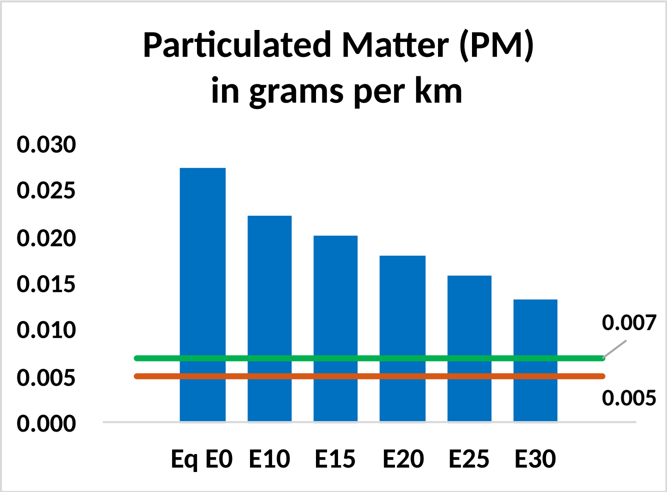












Source: Faro90

Japan – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	7,156,775	6,694,533	6,232,291	5,870,662	5,514,644	5,243,270	4,922,158	-13%	-23%
CO2	396,805,605	386,602,033	376,965,325	369,386,338	365,635,958	363,765,372	357,060,131	-5%	-8%
VOC	871,141	828,487	785,832	755,623	727,407	706,369	675,968	-10%	-16%
NOx	624,372	576,503	544,091	527,804	501,316	480,670	456,287	-13%	-20%
SOx	4,495	4,155	3,815	3,528	3,252	2,953	2,639	-15%	-28%
NH3	139,266	137,887	136,508	137,156	138,699	139,785	140,686	-2%	0%
N2O	8,599	8,552	8,518	8,559	8,736	8,833	8,933	-1%	2%
CH4	196,321	196,321	196,321	199,266	203,585	207,315	210,848	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	6,215	5,969	5,724	5,568	5,432	5,333	5,180	-8%	-13%
Acetaldehyde	14,149	16,239	23,827	36,286	49,436	56,490	66,749	68%	249%
Formaldehyde	48,367	47,024	54,816	64,366	67,433	74,599	81,209	13%	39%
Benzene	14,570	14,001	13,267	13,057	12,948	12,383	11,983	-9%	-11%
PM2.5	23,127	21,025	18,922	17,072	15,262	13,321	11,290	-18%	-34%
PM10	28,296	25,723	23,151	20,887	18,673	16,298	13,813	-18%	-34%

Source: Faro90

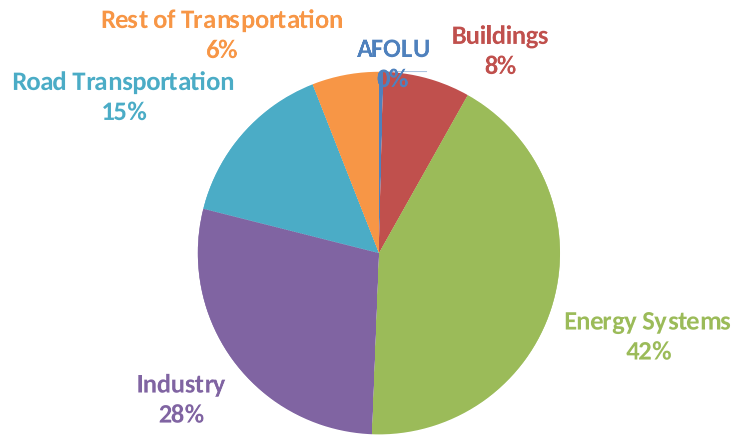
Life Cycle GHG Emissions

The Global Agenda for Climate Action calls for countries, cities and regions, businesses, and members of civil society around the world to to achieve a low-carbon and climate-resilient economies in support of the Paris Agreement. Road Transportation accounts for 15% of total GHG emissions (IPCC,2023).

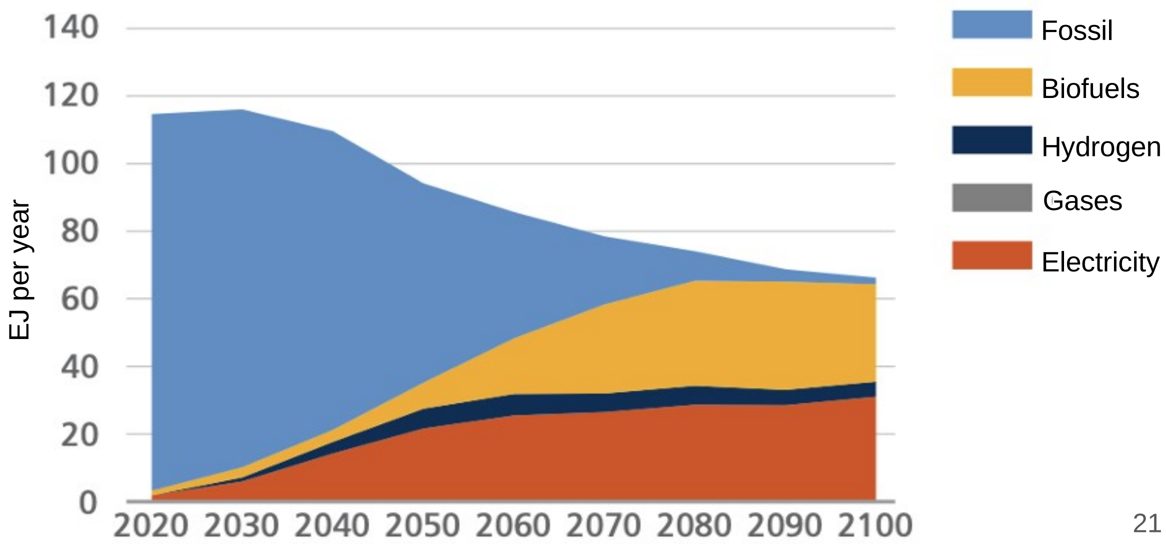
Ethanol gasoline blends are an internationally adopted practice that offer significant short-term benefits. Biofuels are an important solution to achieve reduction targets in the transportation sector.

Country GHG emission reductions are calculated using **RED II/III** and **GREET** life cycle emission methodologies and the **International Vehicular Emissions Model (IVE).** Results are compared to the current country targets for 2035. Current 2025 country emission are those published by **IPCC**.

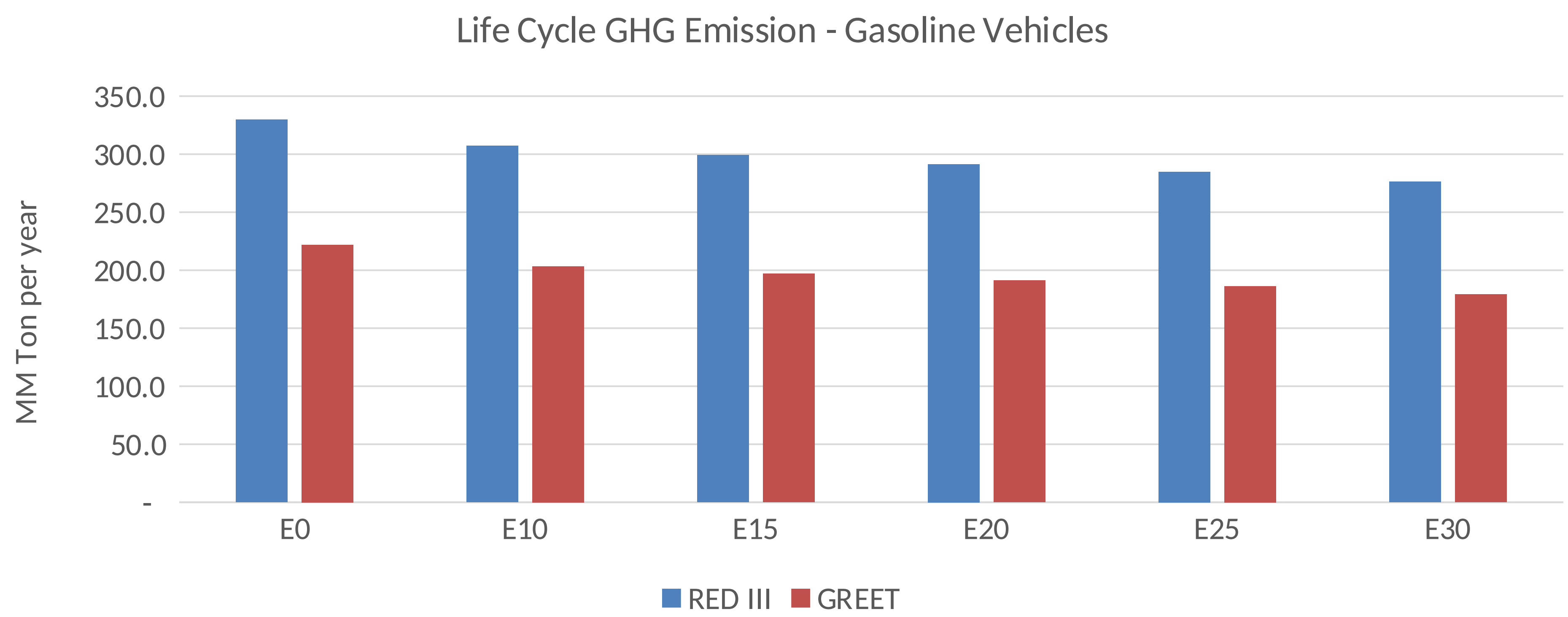
Current GHG Global Emissions Distribution



IPCC Transport Sector Sustainable Development Goals and climate policies Scenario, 2023



Japan – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Country Emissions 2024 (MMT/year)	1,102.2
Target @ 2035 (MMT/year)	570.4
Delta	531.8

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	8.3%	11.2%	13.9%	16.1%	19.2%
RED II/III	0.0%	6.8%	9.3%	11.6%	13.6%	16.2%

% 2035 Target	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	3.5%	4.7%	5.8%	6.7%	8.0%
RED II/III	0.0%	4.2%	5.7%	7.2%	8.4%	10.1%

Source: Greet, RED II/III, climateactiontracker.org, F90